

Operators manual for Attachment



Working with the bucket

Working with the backhoe

Working with the rear loader

Working with the front loader

Manufacturer:

AVANT[®]
AVANT TECNO OY
e-mail: sales@avanttecno.com

Ylötie 1
FIN-33470 YLÖJÄRVI
FINLAND
Tel. +358 3 347 8800
Fax +358 3 348 5511

www.avanttecno.com

1. Introduction

Avant Tecno Oy wants to thank you for purchasing this original Avant attachment. It is the result of Avant's long experience in design and manufacturing of attachments. We ask you that you read and understand the contents of this manual completely before operating the attachment. This will improve your operating and maintenance efficiency, help avoid breakdowns and damage and extend your machines life. Do not hesitate to contact your local Avant dealer for advice regarding service, parts and any problems relating to the operation of your attachment.

2. Designed purpose

Avant heavy-duty silage forks are intended for distributing silage hay, or transporting bales over short distances. The silage forks are well suited for loosening silage hay from silo and then distributing it to the cattle. The silage forks are also suitable for transporting different kinds of round or square bales only limitation being the weight of the bale versus the tipping load of the loader. Be sure not to lift too heavy bales when using the silage forks.

A special cleaning plate is available as an optional extra. The cleaning plate is intended for cleaning the feeding table from left over hay before distributing new hay. Instructions on how to use the plate are written in paragraph 5.3.

The heavy-duty silage forks are intended for use with 300 and 500 series loaders only.

3. Safety instructions



- Never use silage forks without instructions. Read all the decals and this manual.
- Start the operation slowly and carefully.
- Do not wear loose clothing, long uncovered hair or jewellery near the machine.
- When coupling the attachment, make sure that the attachment locking pins lock in positively.
- Keep hands, feet and clothing away from any moving part. Especially from the jaw of silage forks.
- Make sure, that there are no bystanders in the working area before operating the silage forks. **The safety area is 3 meters or 9 feet.**
- Do not leave the cockpit while the loader's engine is running.
- Never go under an uplifted attachment.
- Always carry the attachment **as low as possible**, because of the optimal weight distribution.
- Never lift a too heavy load with the silage forks. Tipping of the loader may cause you to loose control of the loader.
- Never perform **any** service operations when the engine is running.

4. Technical specification

Avant product number	Width (mm)	Length (mm)	Height (mm)	Weight (kg)
A2868	800	770	630	90
A31512	900	770	630	100
A31519	1050	770	630	120

- Heavy-duty silage forks are equipped with forged lower spikes, which are fully changeable.
- All silage forks are equipped with hydraulic jaw compression.

Optional extras include:

- Quick attach cleaning plate (A2936)

5. Attaching silage forks to the loader

5.1 Mechanical attachment

Coupling the attachment to a 300- or 500 series loader happens as follows:

Stage 1:

- Lift up the two locking pins on either side of the attachment coupling plate and turn them backward so that they remain in the locked up position.
- Make sure, that the pins remain in the up position; otherwise you cannot couple the attachment properly!



Stage 2:

- Turn the attachment coupling plate with the tilt movement so that the upper edge of the plate leans forward.
- Drive the loader into the attachment.



Stage 3:

- Lift the boom a little so that the attachment lifts off of the ground.
- Pull the boom control lever to the left so that the lower edge of the attachment coupling plate turns into the attachment.

**Stage 4:**

- Pull the boom control lever more to the left so that attachment coupling plate turns more and the bolts on the boom push the pins down in the holes of the attachment.
- **CAUTION! Make sure that the pins lock properly down in the holes of the attachment.**



Always remember the safety instructions from paragraph 3, when attaching silage forks.

5.2 Connecting the hydraulic couplings

Connecting the hoses to silage fork

With the silage fork follows a plastic bag, which contains two (2) hoses, two (2) hydraulic connectors, two (2) quick couplings and the clamps for binding the hoses. The other end of the hydraulic hoses, which contain a so-called banjo connector are supposed to be attached to the cylinder of silage fork (place shown with arrows in **picture 3**) with the provided special screw see **picture 1**.



Picture 1

Before attaching the connectors to the cylinder make sure that both of the seals are in place. One belongs on top of the connector and one on the bottom of the connector. Otherwise the connection will leak oil during regular usage.

The quick couplings on the other end are supposed to be tightened so that they do not leak during usage. When they are tightened put the protective rubber cap onto the quick coupling. Always put these rubber caps onto the quick coupling when they are not connected to the loader's quick couplings. This prevents the dirt from going into the coupling and thus lengthens lifespan of the quick coupling.

After you have mounted the hoses to the cylinder you must bind the hoses so that they do not get clamped between the boom and wheel of the loader. For binding the hoses you can use the supplied hose clamp shown in **picture 2**.



Picture 2

The clamp works by sliding the hoses through the two half circles of the clamp. When the hoses are between the two plastic parts put the metal cover on top and finally tighten the screw to the designated place circled in **picture 3**.



Picture 3

Coupling the 300 and 500 series loader

The silage forks have two hydraulic quick couplings, which should be attached to the corresponding couplings in the loader. There are two male couplings in the silage fork and correspondingly the loader has two female couplings.

The hoses from the silage fork should be attached to the loader correctly (pressure line to pressure line and return line to return line). The blue rubber caps of the couplings always mark the pressure line (one on the loader and one on the attachment). Always connect them as a pair. This way the silage fork jaw will always operate in the same direction when the auxiliary hydraulics lever is moved.



If you feel that the jaw operates illogically when the loader's auxiliary hydraulics control lever moved, please change the coupling of the return- and pressure hose. **See picture 4**



Picture 4



Make sure that the hydraulic hoses **do not get overstretched** in any boom or attachment plate position and are **not in a position** where they can be **trapped during the operation** of the machine and attachment.

- For coupling the hydraulic hoses see also the operators manual of the loader.
- Always remember the safety instructions from paragraph 3, when coupling the attachment.

5.3 Mounting and demounting the cleaning plate (optional extra)

With the cleaning plate you can easily clean the feeding table of a cow house or similar flat surfaces from left over hay and other debris. The cleaning plate attaches directly to the silage forks with the hydraulic jaw, this way the driver does not even have to get out of the cockpit for using the cleaning plate. Instructions on how to mount the cleaning plate are below.

1. Open the jaw of the silage forks and drive to the backside of the cleaning plate. **See picture 5.**
2. Drive forward and aim the center spike to the slot on the center of the cleaning plate. Make sure that the lower spikes are just a little of the ground.
3. When the lower spikes of the silage forks hit the back wall of the cleaning plate, start to close the jaw of the silage forks as shown in **picture 6**. Make sure that the cleaning plate is properly locked in before using it.



Picture 5



Picture 6

6. Instructions for use

The silage forks are well suited for transporting silage hay, manure and other corresponding material. With a 300 series loader the silage forks jaw is best operated with the foot pedal of auxiliary hydraulics control.

Handling round bales

With most Avant loaders you are able to lift and transport round bales. The loader models which are able to lift a round bale are 320, 320+, 520, 520+, 523+, 528+ and

530+. Avant Tecno cannot guarantee that a loader is able to lift the bale, because different balers make different size and weight bales. In general we can state that the mentioned Avant loaders are able to lift over 90% of the bales on market. For lifting a round bale you have to apply a certain technique in lifting, which will be explained in the following. If the rear end of the loader raises when applying this technique you can add some rear weights to the back of the loader. The rear weights weigh 36 kg a piece and you can add up to three back weights to a loader simultaneously. The rear weights stabilize the loader remarkably and enable higher lifting capacity. The Avant product number for rear weight is A43428.

The lifting technique

1. Lift the silage forks lower spikes to a mid level of the bale. Make sure that the spikes are in level with the ground and open the jaw of silage forks.
2. Gain some speed with the loader and drive the spikes trough the **end** of the bale.
3. If the lower spikes of the silage forks are deep enough in the bale close the jaw and start lifting the bale. When lifting the bale tip the front edge of the silage forks down wards. When the bale is lifted as instructed the weight of the bale is as close to the loader as possible as described in **picture 7**.



Picture 7

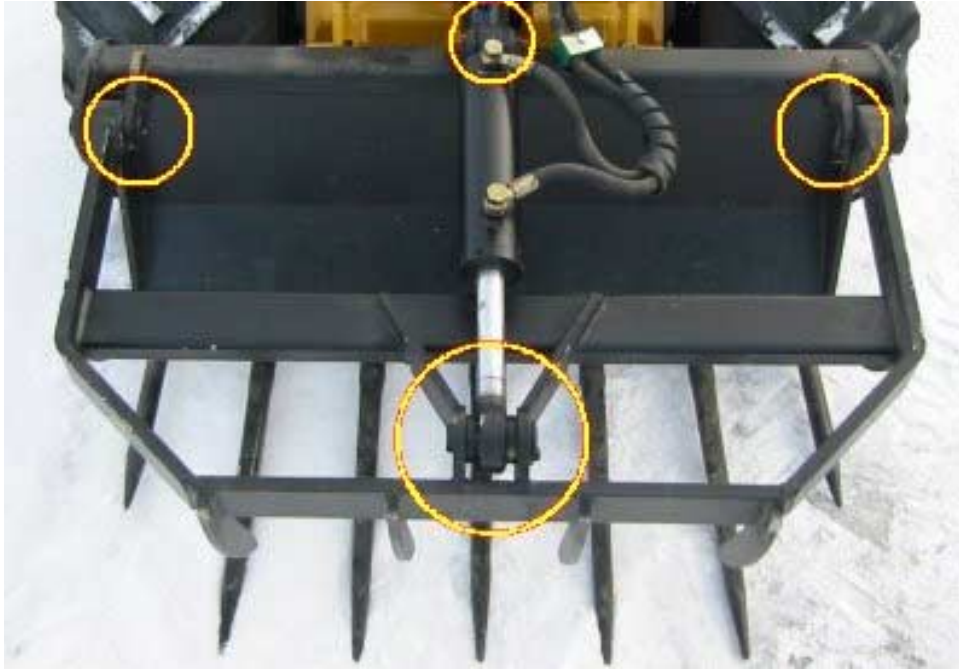


When lifting bales and other heavy loads bear in mind that the center of gravity is much higher than usual. Avoid going into rough terrain and steep hills, because the loader will tip over easier than when the silage forks are empty.

7. Service

7.1 Greasing points

For greasing you need a grease press and a tube of grease, which will fit to the press. Both of these accessories are available at an Avant dealer or a regular hardware store. The silage fork has four (4) greasing points. Two at both ends of the jaw cylinder and two at the pivot points of the jaw. **See picture 8.**



Picture 8

These greasing points should be greased every 10-15 using hours.

Note: Add only couple of presses from the grease press per one greasing. Over greasing will not prevent wear of the pivoting pins.

9. Warranty terms for Avant silage forks (Product numbers A2868, A31512 and A31519)

During the first year of operation Avant Tecno Oy warrants to replace any part or repair any defect, which may occur, subject to the terms below:

1. The product has received regular maintenance in accordance with schedules given by the manufacturer.
2. Any damage caused by operation in a negligent manner or exceeding the approved specifications detailed in this manual is excluded.
3. Avant Tecno Oy accepts no responsibility for interruption to working or any other consequential losses resulting from any failure of the product.
4. Only Avant Tecno Oy approved replacement or original quality parts shall be used during routine maintenance.

In the event a fault occurs which is attributable to manufacturing or assembly defect you should arrange to return your attachment to your authorized dealer for repair. Travel and freight costs are excluded.